

Changing face shape

Just as the cartilage of the voice box is sensitive to testosterone, so, too, are the facial bones. As the facial bones increase in size, they cause air-filled spaces—known as cavities—in the skull to increase in size, too. Larger cavities give the air molecules that have rushed through the vocal cords more space to bounce around, or resonate, deepening the voice further.

▷ Bigger cavities

The different-size cavities within a person's facial structure create the different tones and pitches of the voice. Each person has different-size cavities, which is why everyone has different sounding voices.

The larynx resonates to a small degree, mainly when a person is making higher-pitched sounds.

The frontal cavities are between the eyes, at the top of the nasal bridge.

The sphenoid cavity is deep in the skull, behind the eyes.

Large cavities are found behind the cheekbones.

Squeaks and croaks

During puberty, the changes to males' voices usually start once the penis has finished its rapid growth spurt. Voice changes are usually gradual, as the larynx and facial bones grow relatively slowly. However, in some people, the growth is sudden. With this sudden change can come squeaks and croaks when the teen tries to speak, as the body adjusts to the enlarged voice box. This "breaking voice" can't be controlled, but will pass within a few months, once the larynx stops growing.



TEEN HINTS

What to expect

- The average age for the voice to change is 15, but it can happen earlier or later—every individual's progress through puberty is different. Voices usually break toward the end of puberty, but most males' voices don't fully mature until they're in their twenties.
- You may feel concerned or embarrassed about the sound of your voice, but it's perfectly normal, and people understand. Remember that many others are going through the exact same thing. It will pass within a few months, or even weeks.
- As the voice box grows, it also tilts slightly, and can sometimes protrude from a person's throat as a so-called Adam's apple (laryngeal prominence). This happens to some teens and not to others.

◁ Highs and lows

The sounds a teen's voice makes while the vocal mechanisms change can waver between high-pitched squeaks and husky croaks.